Subject: PowerCo Case, Proposed Data Requirements and Analytical Approach for Churn Investigation

Hi Estelle,

Following our team discussion regarding PowerCo's customer churn challenges, I have outlined the specific client data requirements and analytical methodology needed to test our core hypothesis, whether price sensitivity is the primary driver behind customer attrition or if underlying factors are at play.

1. Client Data Requirements

To thoroughly analyze churn dynamics and test the price sensitivity hypothesis, we will need the following datasets from PowerCo covering at least the past 2–3 years:

- **Customer Account & Historical Profile Data:**
  - *Identifiers:* Customer ID, geographic location/region, industry/sector type.
  - *Contractual Details:* Sign-up date, contract start/end date, tenure, payment method, renewal history, and current churn flag (status).
  - *Product Portfolio:* Number of active products/services (gas, electricity, bundled packages, green energy add-ons).
- **Pricing & Tariff History:**
  - *Historical Rates:* Energy consumption prices and fixed charges broken down by month/quarter over time.
  - *Off-peak vs. Peak Pricing:* Price variations across peak, mid-peak, and off-peak periods.
  - *Competitor & Market Benchmark Data:* Historical average market rates or competitor promotional pricing (if available).
- **Consumption & Usage Metrics:**
  - *Historical Volume:* Monthly gas and electricity consumption figures (kWh/m^3).
  - *Usage Patterns:* Seasonal fluctuations, peak vs. off-peak ratio and sudden drops/spikes in consumption.
- **Customer Interactions & Experience Data:**
  - *Customer Service Touchpoints:* Number of inbound calls, complaints logged, resolution times and channel of contact.
  - *Satisfaction Indicators:* Net Promoter Score (NPS), CSAT ratings or account representative notes.

2. Analytical Approach & Methodology

We will structure our investigation using a systematic 5-step data science process:

1. **Data Ingestion & Cleaning:**
   - Validate data types, handle missing values, correct anomalies and standardize temporal fields (e.g., contract dates).
2. **Exploratory Data Analysis (EDA):**

    - Examine baseline churn rates across customer segments, tenure bands and geographic regions.
    - Evaluate descriptive statistics and distribution of price changes preceding customer churn events.
3. **Feature Engineering:**
    - *Price Sensitivity Metrics:* Calculate price changes over 3, 6 and 12-month rolling windows, price elasticity proxies and relative difference between peak and off-peak tariffs.
    - *Behavioral Indicators:* Usage volatility, ratio of off-peak to peak consumption and complaint frequency scores.
    - *Tenure & Contract Features:* Months remaining on contract, total tenure and contract renewal proximity.
4. **Predictive Modeling & Hypothesis Testing:**
    - Train machine learning models (e.g., Logistic Regression, Random Forest, XGBoost) to predict the probability of churn.
    - Assess feature importance techniques (such as SHAP values) to quantify the exact relative weight of price sensitivity versus customer experience and tenure features.
5. **Insights & Strategic Recommendations:**
    - Segment customers into risk categories based on churn probability and price elasticity.
    - Formulate targeted retention strategies (e.g., proactive discount offerings for high-risk/high-value segments vs. service improvements for experience-sensitive customers).

Please let me know if you'd like to adjust any of these data requests before we submit them to PowerCo. Looking forward to diving into the data.

Best regards,

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